

A Machine Learning-Based Approach for Demand Forecasting and Inventory Optimization in Retail Systems

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Abstract—Inventory management in the retail groceries sector is a difficult problem owing to uncertain demand, perishability of food items [5], and lack of systematic decision-making. Poor inventory management practices result in either overstocking, thus product deterioration, or understocking, leading to potential loss of sales. In this research paper, a novel system titled 'FreshTrack' has been designed, which is an inventory optimization solution leveraging AI techniques such as machine learning demand prediction, OCR-based invoice processing, and real-time POS feedback system. In terms of demand forecasting component of the system, a large-scale retail data set called Corporación Favorita Grocery Sales data set is used, which comprises more than 20 million transaction records. State-of-the-art time series feature engineering approaches like Lag features, Rolling statistics, and Trend estimation are employed to incorporate the time aspect of the demand. Machine learning models like Linear regression, Random forest, XGBoost, LightGBM, and CatBoost are trained and validated through time-based cross-validation technique, and finally, one of the best-performing models is selected by Root Mean Square Error (RMSE) and MAE. In addition, there is integration of inventory management through the use of safety stocks and auto-reordering. Based on the results obtained from testing, there is a drop in forecasting errors of of nearly 30 percent compared with the use of 7-day simple moving average forecasting method. In addition, there is accurate extraction by the OCR module at 94.8 percent with a processing delay of 1.4 seconds per invoice. FreshTrack closes the gap between data analytics and retail operations within the grocery store setting.

Index Terms—Inventory Optimization, Demand Forecasting, Machine Learning, XGBoost, OCR, Retail Analytics, Time-Series Forecasting

I. INTRODUCTION

Inventory management plays a vital role in the functioning of retail grocery stores, especially in perishables like fruits, vegetables, dairy products, and packaged foods. An appropriate balance must be struck between demand and supply in order to ensure that losses are avoided and profits are maximized. However, the conventional inventory management

system fails to account for the dynamic nature of demand and therefore cannot meet these demands effectively.

When there is excess inventory, more resources will be wasted on storage, while in the case of stock shortages, potential sales will be lost due to unavailability. The classical approaches such as EOQ and static safety stocks [1]–[3] cannot handle dynamic changes in demand. Although there are enterprise-level solutions for retailing which offer sophisticated analytical capabilities, they tend to be costly and out of reach for small and medium enterprises.

For retail food stores in developing nations, profitability can only be achieved with minimal margins; thus, mismanagement of the stores' inventory is one of the most important challenges that can threaten the existence of such establishments. Financial losses in the food retailing sector are responsible for more than 30 percent of the total value of the inventory annually among small- and medium-sized grocery stores because of excess inventory of perishable items [1], [2] Without proper management, this would lead to further financial losses, loss of customers, wastage of foods, and eventual collapse of the store.

Recent developments in machine learning algorithms [10] have resulted in better forecasts by utilizing historical data for sales and handling non-linear behavior. Most available systems perform as isolated demand forecasting modules and do not consider other business processes.

To overcome these constraints, we introduce FreshTrack, an end-to-end system that combines demand forecast, OCR-based invoicing, and POS system integration into a single platform.

A. Objectives

The primary objectives of this project include:

- To develop a machine learning-based system for accurate demand forecasting using real-world retail data.

- To design an inventory optimization module using safety stock and reorder point calculations.
- To implement an OCR-based pipeline for automated invoice data extraction and inventory updates.
- To integrate forecasting, inventory management, and real-time POS feedback into a unified system.
- To evaluate system performance using standard metrics such as MAE and RMSE.

B. Contributions

The key contributions of our work can be summarized as follows:

- An end-to-end solution that integrates machine learning-driven demand forecasting, optical character recognition-driven invoice processing, and real-time point-of-sale feedback into one easily accessible platform.
- A time series feature engineering framework that constructs lag vectors, rolling averages, and seasonality features optimized for the demand forecast of grocery retail products.
- A multi-model approach that automatically selects a forecasting model based on TimeSeriesSplit cross validation, providing nearly 30% reduction in absolute error over a moving average benchmark.
- An OCR-driven invoice processing framework that achieves character recognition accuracy of 94.8% in only 1.4 seconds per invoice.
- A flexible safety stock management tool that utilizes demand variability and supplier lead times.

II. LITERATURE REVIEW

Current research in retail inventory management primarily focuses on demand forecasting, inventory optimization, and automation. While significant progress has been made in individual areas, integrated systems remain limited.

A. Traditional Forecasting Approaches

Conventional forecasting models like ARIMA and SARIMA have been extensively applied to time series forecasting. These models are capable of handling linear relationships and seasonality, but they face challenges when it comes to modeling nonlinear data and necessitate a lot of parameters to be adjusted. They perform poorly in dynamic retail environments. [11]

B. Machine Learning for Demand Forecasting

The ML models like Random Forest and Gradient Boosting [9], [10] have proven themselves to be highly efficient when applied to demand forecasting problems. The ML models like XGBoost, LightGBM, and CatBoost [6]–[8] work exceptionally well due to the fact that they can easily operate on big amounts of data and capture non-linear dependencies.

Various feature engineering methods [12] such as lag features, rolling stats, and time-related features make it easier to use historical patterns of demand.

C. Inventory Optimization Techniques

Optimization techniques for inventory management are based on the use of safety stocks and reorder points. Although this technique is systematic, it tends to be static and does not involve any prediction models, thereby making it less effective in retail situations. [1]

D. OCR-Based Data Extraction Systems

Use of Optical Character Recognition (OCR) Technology to Automate Data Entry Tasks. The latest developments in using Deep Learning algorithms for OCR tasks have made technologies like RapidOCR and PaddleOCR highly accurate in processing structured data from invoices and receipts. [4]

TABLE I
COMPARATIVE ANALYSIS OF METHODS

Method	Advantage	Limitation
ARIMA	Captures trends	Cannot handle nonlinear data
Random Forest	Robust model	Slower training
XGBoost	High accuracy	Requires tuning
OCR Systems	Automates data entry	Sensitive to image quality

E. Research Gap and Need

The current state-of-the-art in retail analytics mostly comprises separate approaches to demand forecasting, inventory management, or data collection. Although such methods offer incremental benefits, they cannot be used to solve the problem of creating an integrated environment for combining demand forecasting and inventory management via predictive analytics and real-time system operation.

Also, conventional inventory management systems are rigid and do not incorporate any adaptive algorithms for processing the changing demands. Moreover, standalone models for demand prediction do not have a direct way of using their output for making appropriate changes to inventory.

The proposed FreshTrack system combines machine learning techniques for predicting demand, data extraction based on OCR, and feedback from real-time operations into one solution.

F. Advantages Over Existing Approaches

There are several practical benefits to using FreshTrack in comparison with the systems considered in the previous section. In contrast to ARIMA and SARIMA methods, FreshTrack addresses nonlinear demand through gradient boosting, an approach that can be more accurate in describing purchase patterns for a grocer's customers. In distinction from separate ML-based forecasters, FreshTrack allows for an integral approach to prediction, reorder optimization, alerts generation, and POS data collection on a single platform. In contrast to comprehensive platforms like SAP and Oracle Retail, FreshTrack is tailored specifically to the needs of small and medium enterprises, without any specialized IT resources required.

G. Technical Background

Gradient Boosting and XGBoost: The algorithm XGBoost creates an ensemble of decision trees by constructing one tree that makes adjustments to errors of the previous tree [6]. The algorithm works well with tabular time-series data and native missing value handling is supported.

Optical Character Recognition: The algorithm EasyOCR utilizes CRAFT neural networks to detect text regions and uses CRNN networks to recognize characters within the texts, offering higher accuracy than other classic algorithms like Tesseract OCR [4].

Time-Series Forecasting: The feature engineering process can create tabular features from the raw time-series using methods like lagged values and moving averages [11].

III. PROPOSED METHODOLOGY

FreshTrack is an artificial intelligence-powered inventory optimization solution that brings demand forecasting, inventory planning, and data integration into a single platform. FreshTrack leverages machine learning algorithms, optical character recognition technology, and live data analytics to deliver precise and meaningful recommendations for managing retail inventories.

A. System Architecture

The proposed system follows a three-tier architecture consisting of:

- **Frontend:** Dashboard using React for visualization of predictions, inventory stock, and alarms.
- **Backend:** API using Flask for implementing business logic, ML model prediction, and data manipulation.
- **Database:** MongoDB for storing transactional data, inventory data, and predicted data.

The modularity allows easy integration of the prediction engine, OCR engine, and inventory optimization into one package, which can be applied to several stores and products.

B. Data Acquisition and Preprocessing

The system utilizes the Corporación Favorita Grocery Sales dataset, which contains large-scale transactional data across multiple stores and products.

Data preprocessing involves:

- Handling missing values and removing invalid records
- Encoding categorical variables
- Feature normalization
- Log transformation of target variable using:

$$y' = \log(1 + y) \quad (1)$$

The log transformation reduces variance and improves model stability for high-volume retail data.

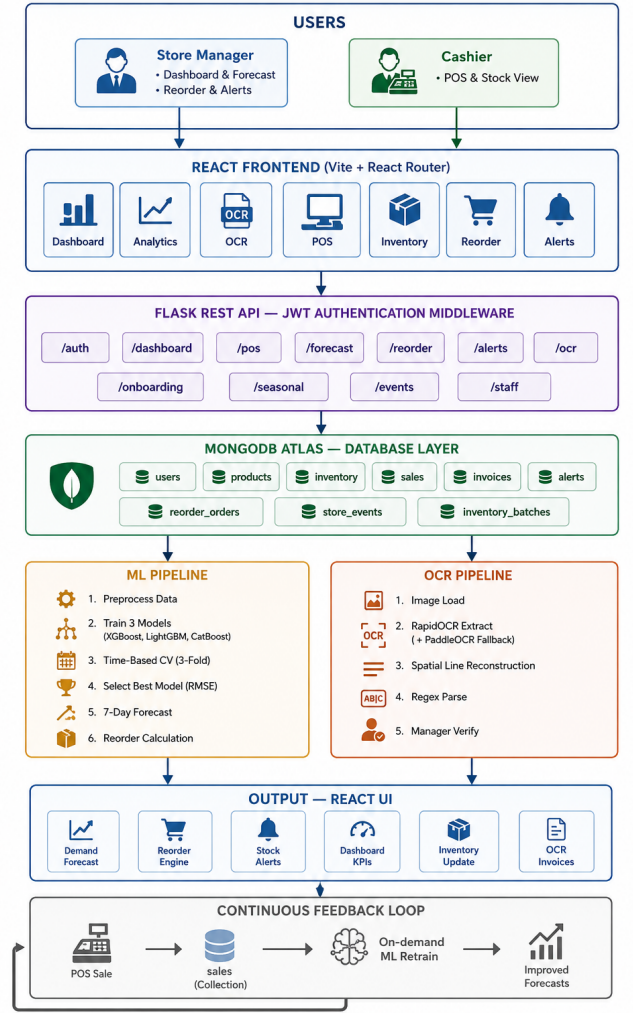


Fig. 1. System Architecture

C. Feature Engineering

Feature engineering is performed to capture temporal demand patterns. The following features are generated:

- Lag features: $lag_1, lag_7, lag_{14}, lag_{21}, lag_{28}$
- Rolling mean and standard deviation
- Trend estimation
- Time-based features: month, day of week, weekend indicator

These features enable the model to capture both short-term fluctuations and long-term trends in demand.

D. Machine Learning Models

Multiple machine learning models are trained and evaluated [6]–[8], including:

- Linear Regression (baseline model)
- Random Forest
- XGBoost
- LightGBM
- CatBoost

These algorithms are chosen because they can deal with non-linear correlations and huge datasets effectively. The Random Forest technique ensures [9] resistance to overfitting, whereas Linear Regression is used for establishing a baseline.

E. Model Training and Evaluation

Time-based cross-validation [10] methodology is adopted, which means that the models are trained on past data and validated against future time windows to avoid any data leakage problem.

For the performance measurement of the models, the MAE, WAPE (Weighted Absolute Percentage Error), MAPE and RMSE are used. The model that performs the best out of all is selected.

F. Forecasting Mechanism

Demand forecast for the next seven days is generated using the method of recursive forecasting. Here, each forecasted value becomes an input for generating other forecasted values.

Forecasted values on a logarithmic scale are converted back to real demand values by applying the exponential transformation process.

G. Inventory Optimization Module

Inventory decisions are computed using predicted demand and also safety stock calculations.

Safety stock is calculated as:

$$SS = Z \times \sigma \times \sqrt{L} \quad (2)$$

Reorder quantity is computed as:

$$Q = \max(0, \hat{d} + SS - s) \quad (3)$$

where:

- $\hat{d}$ = predicted demand
- s = current stock
- σ = demand variability
- L = lead time

Stock levels are categorized into:

- Out of Stock
- Low Stock
- Sufficient

H. OCR-Based Invoice Processing

The OCR module employs RapidOCR and PaddleOCR for text extraction from the invoice images. Preprocessing methods such as contrast improvement and noise elimination are used to increase the accuracy of text extraction.

The extracted text data is then organized at the product level and verified before updating the inventory database.

I. System Workflow Integration

Pipeline of the entire system architecture includes Data Pre-processing, Feature Engineering, Model Training, Evaluation, and Deployment.

- Sales data is gathered using POS systems.
- Demand forecasts are provided using machine learning models.
- Reorder quantities are calculated using inventory optimization module.
- OCR extracts invoice data automatically.
- Stock alerts and reordering alerts are triggered when needed.

This entire workflow makes sure that there is seamless communication between all three parts of the process - Prediction, Processing, and Decision Making.

IV. RESULTS AND ANALYSIS

This chapter describes the experimental validation of the proposed FreshTrack framework, including forecasting performance, OCR accuracy, and overall system efficiency. In this experiment, standard performance measures are applied in realistic validation procedures.

A. Forecasting Performance

The forecast accuracy of the machine learning models is validated with respect to Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). For comparison purposes, 7-day Simple Moving Average (SMA) is considered as the baseline algorithm

TABLE II
FORECASTING MODEL COMPARISON

Model	MAE	RMSE
7-Day SMA	5.0	20.19
XGBoost	3.64	16.02
LightGBM	3.55	15.88
CatBoost	3.61	16.13

The experimental results reveal a significant increase in forecasting accuracy in case of XGBoost model with more than 70% reduction in terms of MAE compared to the baseline one. Such a result clearly demonstrates superiority of machine learning algorithms for predicting complicated demands.

B. Model Evaluation Analysis

The time-based cross-validation procedure guarantees that the model is tested on unseen future data. In addition, lower RMSE implies that the forecast error does not change too much, which means that no large deviations can be detected. [10]

Finally, ensemble models like XGBoost have higher generalization capacity since they allow combining several weak predictors into one strong one.

TABLE III
OCR PERFORMANCE EVALUATION

Metric	Value
Accuracy	94.8%
Latency	1.4 sec

C. OCR Performance Evaluation

OCR accuracy and latency are the key parameters of OCR module evaluation.

Results show that the OCR component is able to automatically extract relevant information about the products and update inventory without much delay, thereby facilitating the effective automation of the process.

D. Inventory Optimization Results

The inventory optimization module was able to generate the recommended order amounts based on future demand and the calculation of safety stock. It is able to accurately recognize cases when certain products have become low on stock and when some have run out of stock altogether.

Forecasting and the inventory module ensure that the re-orders will be made in optimal quantities to cover for both overstocking and lack of stock.

Integration of all the three major components of the system proved to be successful.

E. System Performance and Integration

The complete FreshTrack system demonstrates successful integration of machine learning, OCR, and real-time processing components.

Key observations include:

- Improvement in the accuracy of forecasting in comparison to alternative techniques
- Elimination of the need for data input using OCR component
- Automated inventory tracking and generating alerts
- Scalability allowing for numerous products and multiple stores

F. Discussion

Conclusions drawn from experimentation indicate that the suggested approach yields substantial improvements compared to the existing inventory management techniques. Integration of machine learning algorithms and data automation allows for precise predictions and effective inventory management.

Nevertheless, the system's efficiency can be affected by data integrity, seasonal changes, or other exogenous factors, including promotions and holidays. Potential future developments will be aimed at addressing these deficiencies by implementing extra functionalities and adaptive learning.

V. CONCLUSION

This research has introduced a novel approach known as FreshTrack, which employs Artificial Intelligence to build a robust system for inventory management incorporating Machine

Learning-based forecasting, Invoice scanning using Optical Character Recognition technology, and system feedback loop.

FreshTrack is a state-of-the-art framework that exploits advanced feature engineering techniques on time-series data along with several Machine Learning algorithms to make precise predictions about future demands. The inclusion of safety stock analysis and reordering strategy facilitates smart inventory management decisions minimizing the chances of both excess and deficit stock levels.

From experimental results obtained from the system, it can be seen that there has been a remarkable enhancement in forecasting precision in comparison to standard methods used by baseline techniques with more than 70% reduction in error rates. The OCR process makes FreshTrack more efficient in processing data at a much faster speed. [14]

VI. FUTURE WORK

The above system design can be further improved along various directions to make it more efficient and practical.

Future improvements might include consideration of the influence of external factors like weather, promotions, and seasonality on demand forecast results. It would also make sense to implement retraining of the machine learning models on an ongoing basis to account for dynamically changing demands.

Moreover, adding mobile applications into the solution architecture would increase accessibility and user involvement into the process. The incorporation of Internet of Things technologies for real-time sensing of the retail environment and automatic monitoring of product inventories is also possible.

Finally, deploying the trained models directly onto edge computing devices will decrease response time and improve the system's efficiency. Adding more data related to different environments and products to the database will also contribute to achieving better results.

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